ALGEBRA 1 · UNIT 5: LINEAR FUNCTIONS AND GRAPHING

Mountain Trail Designer

Rate ski trails by slope and design a new run for Pine Ridge Mountain.

Honors

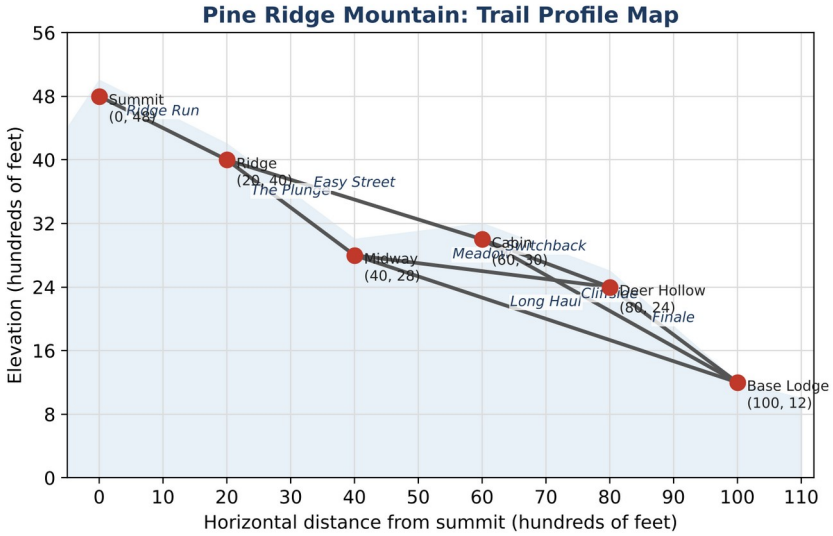
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Pine Ridge Mountain is opening for the season, and the resort needs every trail rated before skiers arrive. Ratings depend on how steep a trail is, and steepness is slope!

The profile map shows each trail as a line segment between two points. The x-coordinate is the horizontal distance from the summit and the y-coordinate is the elevation, both in hundreds of feet. Your job: rate the trails, write their equations, and design a new trail.



Learning Goals

- I can express slope ratings as absolute value inequalities and percent grades.
- I can write equations in slope-intercept, point-slope, and standard form.
- I can design a route by solving inequalities for an unknown coordinate.

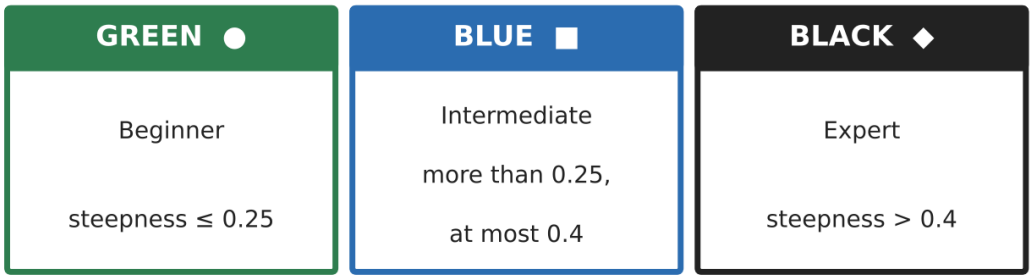
Project Timeline

When	What you will do
Day 1	Ratings as Inequalities
Day 2	Equations & Relationships
Day 3	Engineer an All-Blue Route
Final	New Trail Map + reflection

Part 1: Ratings as Inequalities

Day 1 · Use the map and rating card. Let m be the slope of a trail.

1. Write each rating as an inequality using $|m|$.



Steepness = the slope without its negative sign (its absolute value).

2. Find the slope and rating of every trail. Record them in the table.

Trail	Slope	Steepness	Rating
Ridge Run			
The Plunge			
Long Haul			
Easy Street			
Cliffside			
Meadow			
Switchback			
Finale			

3. Ski resorts often give the grade as a percent: grade = $100|m|\%$. Find the grade of the steepest and least steep trails.

4. Prove that no all-green-or-blue route exists from the Summit to the Base Lodge.

Part 2: Equations & Relationships

Day 2 · Use exact values.

5. Write the equation of the chairlift in slope-intercept form and in standard form Ax + By = C (A, B, and C integers).

6. Show that the Ridge Run line and the Midway-to-Deer Hollow line are not parallel, and find where the two lines (extended) intersect.

7. A snow fence through the Cabin (60, 30) must be perpendicular to Cliffside. Write its equation in standard form.

8. The groomer's elevation is a linear function of time with f(2) = 1,320 and f(14) = 2,040. Write f(t) and find the time when f(t) = 4,800.

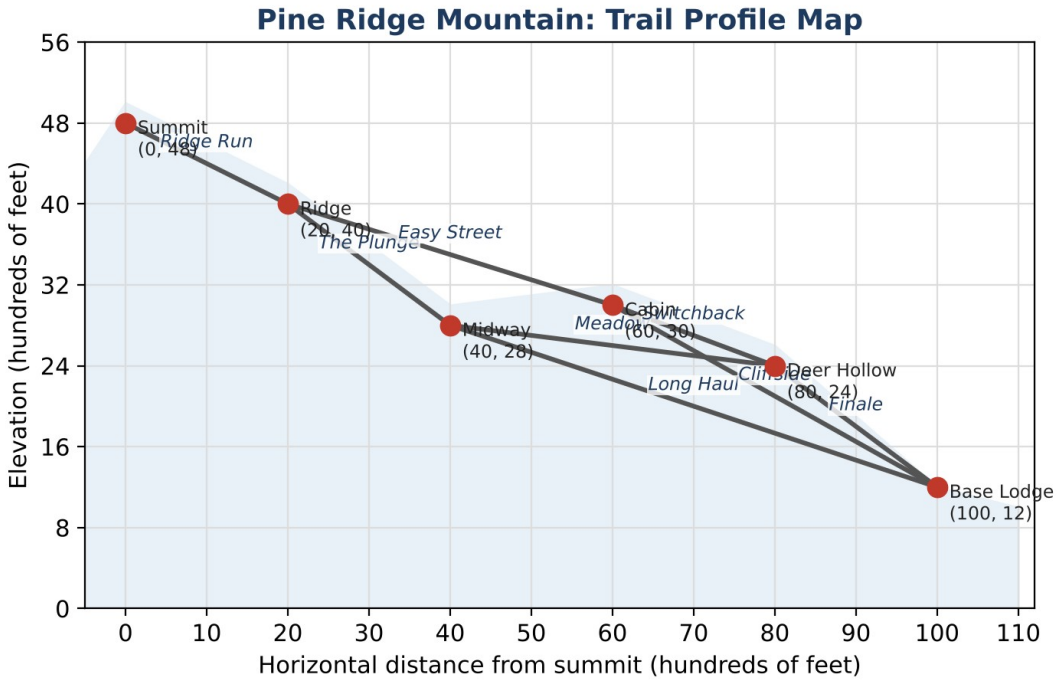
Part 3: Engineer an All-Blue Route

Day 3 · The resort will add a new point E at x = 60 so that Ridge (20, 40) → E → Base Lodge (100, 12) is a route of blue trails.

9. Let E = (60, y). Write a compound inequality for the steepness of Ridge-to-E so that it is blue, and solve for y.

10. Do the same for E-to-Base Lodge.

11. Find all values of y that make both trails blue. Choose one value, write both trail equations, and add the route to your map with a recommendation.



Final Product: New Trail Map

Your final product must include:

- Rating inequalities and the complete slope table with percent grades
- Equations in multiple forms (chairlift, snow fence, groomer)
- The intersection of the two extended trail lines
- The compound inequality for E and your chosen design
- A labeled map with the new all-blue route

Reflection

R1. How did solving for y in Part 3 turn a design problem into an inequality problem?

R2. When is standard form more useful than slope-intercept form?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	New trail meets the rating requirements, with slopes and equations shown. Goes beyond the minimum with a creative, realistic choice.	New trail meets the rating requirements, with slopes and equations shown.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Trail map is labeled with points, slopes, ratings, and equations. Polished, organized, and visually engaging.	Trail map is labeled with points, slopes, ratings, and equations. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.